



Electrodynamics

电动力学

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1 电磁场的基本方程

Maxwell 方程组

$$\begin{cases} \nabla \cdot \mathbf{D} = \rho \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{H} = \mathbf{j} + \frac{\partial \mathbf{D}}{\partial t} \end{cases}$$

本构方程

$$\begin{cases} \mathbf{D} = \varepsilon \mathbf{E} \\ \mathbf{B} = \mu \mathbf{H} \\ \mathbf{j} = \sigma \mathbf{E} \end{cases}$$

边界条件

设 $\mathbf{n}$ 为介质 1 指向介质 2 的单位法向量

$$\begin{cases} \mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma \\ \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0 \\ \mathbf{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0 \\ \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \alpha \end{cases}$$

2 静电场

静电场求解思路

$$\begin{cases} \nabla \cdot \mathbf{D} = \rho \\ \nabla \times \mathbf{E} = 0 \end{cases} \implies \mathbf{E} = -\nabla \varphi$$

$$\mathbf{E}(\mathbf{r}) = \int \frac{\rho(\mathbf{x}')}{4\pi\varepsilon_0 r^2} dV'$$

$$\varphi(\mathbf{r}) = \int \frac{\rho(\mathbf{x}')}{4\pi\varepsilon_0 r} dV'$$

- 对于具有对称性的电场

$$\oint \mathbf{D} \cdot d\mathbf{S} = Q \implies \mathbf{D} \xrightarrow{D=\varepsilon E} \mathbf{E}$$

$$\varphi(r) = \int_r^\infty \mathbf{E} \cdot d\mathbf{l} = \int_r^{R_1} \mathbf{E}_1 \cdot d\mathbf{l} + \int_{R_1}^{R_2} \mathbf{E}_2 \cdot d\mathbf{l} + \int_{R_2}^{R_3} \mathbf{E}_3 \cdot d\mathbf{l} + \dots$$

- 对于复杂问题, 一般要求解 Poisson 方程

$$\nabla^2 \varphi = -\frac{\rho}{\varepsilon}$$

- 当 $\rho = 0$ 时退化为 Laplace 方程

$$\nabla^2 \varphi = 0$$

分离变量法: 轴对称通解

在球坐标下且具有轴对称性时, Laplace 方程 $\nabla^2 \varphi = 0$ 通解的形式为

$$\varphi(r, \theta) = \sum_n \left(a_n r^n + \frac{b_n}{r^{n+1}} \right) P_n(\cos \theta)$$

根据边界条件求出 a_n, b_n

分离变量法: 边界条件

- $r \rightarrow \infty$

$$\begin{cases} \varphi = \varphi_0 - E_0 r \cos \theta & \text{有外场} \\ \varphi = 0 & \text{无外场} \end{cases}$$

- $r \rightarrow 0, \varphi$ 有限

- $r \rightarrow R$ 的介质间分界面

$$\begin{cases} \varphi_1|_{r=R} = \varphi_2|_{r=R} \\ \varepsilon_1 \frac{\partial \varphi_1}{\partial r} \Big|_{r=R} = \varepsilon_2 \frac{\partial \varphi_2}{\partial r} \Big|_{r=R} \end{cases}$$

- $r \rightarrow R$ 的导体表面

$$\begin{cases} \varphi = \varphi_s \\ \oint \sigma dS = Q \quad \sigma = \mathbf{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) \end{cases}$$

镜像法

使用条件: $\begin{cases} \text{电荷数目有限} \\ \text{边界规则} \end{cases}$

镜像电荷的引入, 不能改变边界条件, 且镜像电荷不能在所求空间内. 镜像电荷的电荷量不一定等于边界上的感应电荷量, 但其效应完全等价于感应电荷产生的效应, 如电场、电势、作用力. 镜像电荷可以不止一个.

格林函数法

- 第一类边界条件 $\varphi = C|_S$

$$\varphi(\mathbf{x}) = \int_V \rho(\mathbf{x}') G(\mathbf{x}', \mathbf{x}) dV' - \varepsilon_0 \oint_S \varphi(\mathbf{x}') \frac{\partial G(\mathbf{x}', \mathbf{x})}{\partial n'} dS'$$

- 第二类边界条件 $\left. \frac{\partial \varphi}{\partial n} \right|_S = C$

$$\varphi(\mathbf{x}) = \int_V \rho(\mathbf{x}') G(\mathbf{x}', \mathbf{x}) dV' + \varepsilon_0 \oint_S G(\mathbf{x}', \mathbf{x}) \frac{\partial \varphi(\mathbf{x}')}{\partial n'} dS' + \langle \varphi \rangle_S$$

- $G(\mathbf{x}, \mathbf{x}')$ 为 Green 函数, 满足

$$\nabla^2 G(\mathbf{x}, \mathbf{x}') = -\frac{1}{\varepsilon_0} \delta(\mathbf{x} - \mathbf{x}')$$

电多极展开

$$\varphi(\mathbf{x}) = \frac{Q}{4\pi\varepsilon_0 r} + \frac{\mathbf{p} \cdot \mathbf{r}}{4\pi\varepsilon_0 r^3} + \frac{1}{4\pi\varepsilon_0} \frac{1}{6} \mathcal{D} : \nabla \nabla \frac{1}{r} + \dots$$

- 总电荷

$$Q = \int_V \rho(\mathbf{x}') dV'$$

- 电偶极矩

$$\mathbf{p} = \int_V \rho(\mathbf{x}') \mathbf{x}' dV'$$

- 电四极矩张量

$$\mathcal{D} = \int_V 3\mathbf{x}' \mathbf{x}' \rho(\mathbf{x}') dV'$$

电偶极矩在外场中的能量

$$W = \mathbf{p} \cdot \nabla \varphi_e = -\mathbf{p} \cdot \mathbf{E}_e$$

电四极矩在外场中的能量

$$W = -\frac{1}{6} \mathcal{D} : \nabla E_e$$

3 静磁场

静磁场基本方程

$$\begin{cases} \nabla \times \mathbf{H} = \mathbf{J} \\ \nabla \cdot \mathbf{B} = 0 \end{cases}$$

边值关系

$$\begin{cases} \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \boldsymbol{\alpha} \\ \mathbf{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0 \end{cases}$$

对称磁场的求解思路

由 Ampère 环路定理

$$\oint \mathbf{H} \cdot d\mathbf{l} = I \Rightarrow \mathbf{H} \Rightarrow \mathbf{B} = \mu \mathbf{H}$$

磁矢势

由 $\nabla \cdot \mathbf{B} = 0$ 引入矢势

$$\mathbf{B} = \nabla \times \mathbf{A} \quad \mathbf{A} = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{x}')}{r} dV'$$

磁标势

当 $\mathbf{J} = 0$ 且 $\oint \mathbf{H} \cdot d\mathbf{l} = 0$ (单连通区域) 时, 可引入磁标势

$$\mathbf{H} = -\nabla \varphi_m$$

$$\nabla^2 \varphi_m = -\frac{\rho_m}{\mu_0} \quad (\rho_m = -\mu_0 \nabla \cdot \mathbf{M})$$

当 $\rho_m = 0$ 时退化为 Laplace 方程

$$\nabla^2 \varphi_m = 0$$

通解

与电势一样, 在球坐标下且具有轴对称性, 有通解

$$\varphi_m(r, \theta) = \sum_n \left(a_n r^n + \frac{b_n}{r^{n+1}} \right) P_n(\cos \theta)$$

边界条件

$$\begin{cases} \varphi_{m1}|_S = \varphi_{m2}|_S \\ \mu_1 \frac{\partial \varphi_{m1}}{\partial n}|_S = \mu_2 \frac{\partial \varphi_{m2}}{\partial n}|_S \end{cases}$$

$$\begin{cases} r \rightarrow 0 & \varphi_m \text{ 有限} \\ r \rightarrow \infty & \begin{cases} \varphi_m = -H_0 r \cos \theta & \text{有外场} \\ \varphi_m = 0 & \text{无外场} \end{cases} \end{cases}$$

矢势矢势 $\mathbf{A}$ 的多极展开

$$\begin{aligned} \mathbf{A}(\mathbf{x}) &= \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{x}')}{r} dV' \\ &= \frac{\mu_0}{4\pi} \int_V \mathbf{J}(\mathbf{x}') \left(\frac{1}{R} - (\mathbf{x}' \cdot \nabla) \frac{1}{R} + \frac{1}{2} (\mathbf{x}' \cdot \nabla)^2 \frac{1}{R} + \dots \right) dV' \\ &= \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{R}}{R^3} + \dots \end{aligned}$$

磁矩

$$\mathbf{m} = \frac{I}{2} \oint (\mathbf{x}' \times d\mathbf{l}) \quad (\mathbf{m} = I\mathbf{S})$$

AB 效应

在量子力学框架下, 矢势 $\mathbf{A}$ 具有可测量的物理意义. 即使在磁场为零的区域, 矢势仍可影响带电粒子的量子相位.

超导体

- 超导体的两大基本特性
 1. 零电阻;
 2. 完全抗磁性: Meissner 效应.
- 临界磁场, 临界电流, 磁通量子化.

- 临界磁场与温度的关系

$$H_c(T) = H_c(0) \left[1 - \left(\frac{T}{T_c} \right)^2 \right] \quad (T \leq T_c)$$

London 方程

London 第一方程

$$\frac{\partial \mathbf{J}_s}{\partial t} = \alpha \mathbf{E} \quad \alpha = \frac{n_s e^2}{m}$$

London 第二方程

$$\nabla \times \mathbf{J}_s = -\alpha \mathbf{B}$$

4 电磁场与物质相互作用

极化强度

$\mathbf{p}_i = q\mathbf{l}$ 为单个分子的电偶极矩.

$$\mathbf{P} = \frac{\sum \mathbf{p}_i}{\Delta V}$$

对各向同性均匀介质

$$\mathbf{P} = \chi_e \varepsilon_0 \mathbf{E}$$

极化电荷体密度

$$\rho_p = -\nabla \cdot \mathbf{P} \quad Q_p = -\oint \mathbf{P} \cdot d\mathbf{S}$$

极化面电荷密度

$$\sigma_p = -\mathbf{n} \cdot (\mathbf{P}_2 - \mathbf{P}_1)$$

介质中的 Gauss 定理

$$\oint \mathbf{D} \cdot d\mathbf{S} = \int \rho dV \quad \nabla \cdot \mathbf{D} = \rho$$

电位移矢量

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}$$

对各向同性均匀介质

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P} = \varepsilon_0 \mathbf{E} + \chi_e \varepsilon_0 \mathbf{E} = \varepsilon_0 (1 + \chi_e) \mathbf{E} = \varepsilon_0 \varepsilon_r \mathbf{E} = \varepsilon \mathbf{E}$$

$\varepsilon_r = 1 + \chi_e$ 相对介电常数, $\varepsilon = \varepsilon_0 \varepsilon_r$ 介电常数.

极化电流

对交变电场, 极化电流密度

$$\mathbf{j}_p = \frac{\partial \mathbf{P}}{\partial t}$$

磁化强度

$\mathbf{m}_i = i\mathbf{a}$ 为单个分子的磁矩

$$\mathbf{M} = \frac{\sum \mathbf{m}_i}{\Delta V}$$

对各向同性均匀介质

$$\mathbf{M} = \chi_m \mathbf{H}$$

磁化电流密度

$$\mathbf{j}_m = \nabla \times \mathbf{M}$$

磁化面电流密度

$$\boldsymbol{\alpha}_m = \mathbf{n} \times (\mathbf{M}_2 - \mathbf{M}_1)$$

介质中的 Ampère 环路定理

$$\oint \mathbf{H} \cdot d\mathbf{l} = \int \mathbf{J} \cdot d\mathbf{S} + \int \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

磁场强度

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M} = \frac{\mathbf{B}}{\mu}$$

Coulomb 相互作用

$$\mathbf{F} = \frac{q_1 q_2}{4\pi \varepsilon_0} \frac{\mathbf{r}}{r^3}$$

场对电荷的相互作用: Lorentz 力 (密度)

$$\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$$

$$\mathbf{f} = \rho\mathbf{E} + \mathbf{J} \times \mathbf{B}$$

5 电磁场的能量和动量

能量密度

$$w = \frac{1}{2}(\mathbf{E} \cdot \mathbf{D} + \mathbf{H} \cdot \mathbf{B})$$

$$\begin{cases} w = \frac{1}{2}\varepsilon E^2 + \frac{1}{2}\frac{B^2}{\mu} & \text{均匀介质} \\ w = \frac{1}{2}\varepsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0} & \text{真空} \end{cases}$$

能流密度 (Poynting 矢量)

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

$$\langle \mathbf{S} \rangle = \frac{1}{2} \text{Re}\{\mathbf{E}^* \times \mathbf{H}\}$$

损耗功率

$$\langle P \rangle = \frac{1}{2} \text{Re}\{\mathbf{J}^* \cdot \mathbf{E}\}$$

动量密度与动量流密度

$$\mathbf{g} = \mathbf{D} \times \mathbf{B}$$

$$\mathcal{T} = -(\mathbf{D}\mathbf{E} + \mathbf{B}\mathbf{H}) + \frac{1}{2}\mathcal{I}(\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H})$$

能量守恒

$$\frac{\partial w}{\partial t} + \mathbf{f} \cdot \mathbf{v} = -\nabla \cdot \mathbf{S}$$

动量守恒

$$\mathbf{f} + \frac{\partial}{\partial t}\mathbf{g} = -\nabla \cdot \mathcal{T}$$

电荷守恒

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0$$

6 电磁波的传播

时谐电磁波

$$\mathbf{E}(\mathbf{x}, t) = \mathbf{E}(\mathbf{x})e^{-i\omega t} \quad \mathbf{B}(\mathbf{x}, t) = \mathbf{B}(\mathbf{x})e^{-i\omega t}$$

Helmholtz 方程

$$\begin{cases} \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} \\ \nabla \cdot \mathbf{D} = 0 \\ \nabla \cdot \mathbf{B} = 0 \end{cases} \xrightarrow[k=\omega\sqrt{\mu\epsilon}]{\frac{\partial_t \rightarrow -i\omega}{k=\omega\sqrt{\mu\epsilon}}} \begin{cases} \nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0 & (\nabla \cdot \mathbf{E} = 0) \\ \nabla^2 \mathbf{B} + k^2 \mathbf{B} = 0 & (\nabla \cdot \mathbf{B} = 0) \end{cases}$$

• 求 $\mathbf{E}$

$$\begin{cases} \nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0 & (\nabla \cdot \mathbf{E} = 0) \\ \mathbf{B} = -\frac{i}{\omega} \nabla \times \mathbf{E} \end{cases}$$

• 求 $\mathbf{B}$

$$\begin{cases} \nabla^2 \mathbf{B} + k^2 \mathbf{B} = 0 & (\nabla \cdot \mathbf{B} = 0) \\ \mathbf{E} = \frac{i}{\omega\mu\epsilon} \nabla \times \mathbf{B} \end{cases}$$

平面电磁波

$$\mathbf{E}(\mathbf{x}, t) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \quad \mathbf{B}(\mathbf{x}, t) = \mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

1. 平面电磁波是 TEM 波, $\mathbf{k} \perp \mathbf{E}$, $\mathbf{k} \perp \mathbf{B}$;
2. $\mathbf{E} \perp \mathbf{B}$, $\mathbf{k} \parallel \mathbf{E} \times \mathbf{B}$, 三者满足右手螺旋关系;
3. $\mathbf{E}$ 与 $\mathbf{B}$ 同相位, 振幅比为相速度 v ;
4. 电磁波传播速度 $v = 1/\sqrt{\mu\epsilon}$.

相速度与群速度

$$v_p = \frac{\omega}{k} \quad v_g = \frac{d\omega}{dk}$$

圆偏振

$$\begin{cases} \mathbf{E} = E_0 e^{i(kz - \omega t)} (\mathbf{e}_x - i\mathbf{e}_y) & \text{右旋} \\ \mathbf{E} = E_0 e^{i(kz - \omega t)} (\mathbf{e}_x + i\mathbf{e}_y) & \text{左旋} \end{cases}$$

反射和折射定律

$$\theta = \theta' \quad n_1 \sin \theta = n_2 \sin \theta''$$

Fresnel 公式

$$\left\{ \begin{array}{ll} \frac{E'_\perp}{E_\perp} = -\frac{\sin(\theta - \theta'')}{\sin(\theta + \theta'')} & \frac{E''_\perp}{E_\perp} = \frac{2 \cos \theta \sin \theta''}{\sin(\theta + \theta'')} \\ \frac{E'_\parallel}{E_\parallel} = \frac{\tan(\theta - \theta'')}{\tan(\theta + \theta'')} & \frac{E''_\parallel}{E_\parallel} = \frac{2 \cos \theta \sin \theta''}{\sin(\theta + \theta'') \cos(\theta - \theta'')} \end{array} \right\}$$

反射系数与透射系数

$$R = \left| \frac{E'}{E} \right|^2 \quad T = 1 - R$$

Brewster 角

$$\begin{aligned} \theta_B + \theta''_B &= 90^\circ \implies \frac{E''_\parallel}{E_\parallel} = 0 \\ \tan \theta_B &= \frac{n_2}{n_1} \end{aligned}$$

全反射

$$\begin{aligned} \sin \theta &> \sin \theta_c = \frac{n_2}{n_1} \\ k''_z &= i k \underbrace{\sqrt{\sin^2 \theta - (n_2/n_1)^2}}_{=\kappa}, \quad k''_x = k_x = k \sin \theta, \quad \mathbf{E}'' = \mathbf{E}''_0 e^{-\kappa z} e^{i(k''_x x - \omega t)}. \\ \kappa^{-1} &= \frac{\lambda_1}{2\pi \sqrt{\sin^2 \theta - (n_2/n_1)^2}} = \frac{\lambda}{2\pi n_1 \sqrt{\sin^2 \theta - (n_2/n_1)^2}} \\ v_p &= \frac{c}{n_1 \sin \theta} \end{aligned}$$

良导体条件

$$\rho(t) = \rho_0 e^{-\frac{\sigma}{\varepsilon} t}$$

$$\omega \ll \tau^{-1} = \frac{\sigma}{\varepsilon} \implies \frac{\sigma}{\varepsilon \omega} \gg 1$$

复介电常量与复波矢

$$\varepsilon' = \varepsilon + i \frac{\sigma}{\omega}$$

$$k = \omega \sqrt{\mu \varepsilon'} \quad \mathbf{k} = \beta + i \alpha$$

趋肤深度

$$\delta = \frac{1}{\alpha} \approx \sqrt{\frac{2}{\omega \mu \sigma}}$$

理想导体壁边界条件

$$\begin{cases} \mathbf{n} \times \mathbf{E} = 0 \\ \frac{\partial E_n}{\partial n} = 0 \end{cases}$$

矩形谐振腔

$$\begin{cases} E_x = A_1 \cos k_x x \sin k_y y \sin k_z z \\ E_y = A_2 \sin k_x x \cos k_y y \sin k_z z \\ E_z = A_3 \sin k_x x \sin k_y y \cos k_z z \end{cases}$$

$$k_x = \frac{m\pi}{L_1}, \quad k_y = \frac{n\pi}{L_2}, \quad k_z = \frac{p\pi}{L_3} \quad (m, n, p = 0, 1, 2, 3, \dots)$$

由 $\nabla \cdot \mathbf{E} = 0$ 得

$$k_x A_1 + k_y A_2 + k_z A_3 = 0$$

一组 (m, n, p) 对应两种独立的电磁波模式.

谐振频率 (本征频率)

$$\omega_{mnp} = \frac{\pi}{\sqrt{\mu \varepsilon}} \sqrt{\left(\frac{m}{L_1}\right)^2 + \left(\frac{n}{L_2}\right)^2 + \left(\frac{p}{L_3}\right)^2}$$

矩形波导

$$\begin{cases} E_x = A_1 \cos k_x x \sin k_y y e^{i(k_z z - \omega t)} \\ E_y = A_2 \sin k_x x \cos k_y y e^{i(k_z z - \omega t)} \\ E_z = A_3 \sin k_x x \sin k_y y e^{i(k_z z - \omega t)} \end{cases}$$

$$k_x = \frac{m\pi}{a}, \quad k_y = \frac{n\pi}{b} \quad (m, n = 0, 1, 2, 3, \dots)$$

由 $\nabla \cdot \mathbf{E} = 0$ 得

$$k_x A_1 + k_y A_2 - i k_z A_3 = 0$$

一组 (m, n) 对应两种独立模式.

磁场模式:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = i\omega\mu\mathbf{H}$$

波导中只能传播 TE 模或 TM 模 (不能传播 TEM 模).

截止频率与截止波长

$$\omega_{\text{cut}} = \frac{\pi}{\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\lambda_{\text{cut}} = \frac{2ab}{\sqrt{(mb)^2 + (na)^2}}$$

等离子体

Debye 屏蔽 Coulomb 势

$$\varphi(r) = \frac{q}{4\pi\epsilon_0 r} e^{-r/\lambda_D} \quad \lambda_D = \sqrt{\frac{\epsilon_0 kT}{ne^2}}$$

等离子体的振荡频率

$$\omega_p = \sqrt{\frac{ne^2}{m\epsilon_0}}$$

等离子体中电磁波的传播

$$\frac{\partial \mathbf{j}}{\partial t} = \frac{ne^2}{m} \mathbf{E}$$

$$\mathbf{j}(\omega) = \sigma(\omega) \mathbf{E}(\omega) \quad \sigma(\omega) = i \frac{ne^2}{m\omega}$$

$\omega > \omega_p$ 时电磁波才能在等离子体中传播, 故 ω_p 即为截止频率.

7 电磁波辐射

电磁场的矢势和标势

$$\mathbf{B} = \nabla \times \mathbf{A} \quad \mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} - \nabla \varphi$$

常用的规范条件

1. Coulomb 规范

$$\nabla \cdot \mathbf{A} = 0$$

2. Lorenz 规范

$$\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$$

d'Alembert 方程

$$\begin{cases} \nabla^2 \mathbf{A} - \frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu_0 \mathbf{J} \\ \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0} \end{cases} \quad \left(\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0 \right)$$

推迟势

$$\begin{aligned} \mathbf{A}(\mathbf{x}, t) &= \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{x}', t - r/c)}{r} dV' \\ \varphi(\mathbf{x}, t) &= \frac{1}{4\pi\varepsilon_0} \int_V \frac{\rho(\mathbf{x}', t - r/c)}{r} dV' \end{aligned}$$

小区域辐射

$$\text{条件: } l \ll \lambda, l \ll r. \quad \begin{cases} \text{近区 (静态区): } r \ll \lambda, kr \ll 1 \\ \text{感应区 (过渡区): } r \sim \lambda, kr \sim 1 \\ \text{远区 (辐射区): } r \gg \lambda, kr \gg 1 \end{cases}$$

远区辐射

$$\begin{aligned} \mathbf{A}(\mathbf{x}) &= \frac{\mu_0 e^{ikR}}{4\pi R} \int_V \mathbf{J}(\mathbf{x}') \left[1 - ik \mathbf{n} \cdot \mathbf{x}' + \frac{1}{2!} (ik \mathbf{n} \cdot \mathbf{x}')^2 + \cdots \right] dV' \\ &= \mathbf{A}_{(1)}(\mathbf{x}) + \mathbf{A}_{(2)}(\mathbf{x}) + \cdots \end{aligned}$$

电偶极辐射

$$\begin{aligned} \mathbf{A}_{(1)}(\mathbf{x}) &= \frac{\mu_0 e^{ikR}}{4\pi R} \int_V \mathbf{J}(\mathbf{x}') dV' = \frac{\mu_0 e^{ikR}}{4\pi R} \dot{\mathbf{p}} \\ \mathbf{B} = \nabla \times \mathbf{A} &= \frac{e^{ikR}}{4\pi\epsilon_0 c^3 R} \ddot{\mathbf{p}} \times \mathbf{e}_r = \frac{1}{4\pi\epsilon_0 c^3 R} e^{ikR} |\ddot{\mathbf{p}}| \sin\theta \mathbf{e}_\varphi \\ \mathbf{E} = c\mathbf{B} \times \mathbf{e}_r &= \frac{e^{ikR}}{4\pi\epsilon_0 c^2 R} (\ddot{\mathbf{p}} \times \mathbf{e}_r) \times \mathbf{e}_r = \frac{1}{4\pi\epsilon_0 c^2 R} e^{ikR} |\ddot{\mathbf{p}}| \sin\theta \mathbf{e}_\theta \end{aligned}$$

电偶极辐射平均能流密度

$$\mathbf{S} = \frac{|\ddot{\mathbf{p}}|^2}{32\pi^2 \epsilon_0 c^3 R^2} \sin^2\theta \mathbf{e}_r$$

电偶极辐射全空间平均辐射总功率

$$P = \frac{1}{4\pi\epsilon_0} \frac{|\ddot{\mathbf{p}}|^2}{3c^3}$$

短天线的辐射电阻

$$R_r = \frac{\pi}{6} \sqrt{\frac{\mu_0}{\epsilon_0}} \left(\frac{l}{\lambda}\right)^2 = 197 \left(\frac{l}{\lambda}\right)^2 \Omega$$

半波天线电流分布

$$I(z) = \begin{cases} I_0 \sin k \left(\frac{l}{2} - z\right) & 0 \leq z \leq \frac{l}{2} \\ I_0 \sin k \left(\frac{l}{2} + z\right) & -\frac{l}{2} \leq z < 0 \end{cases}$$

半波天线的矢势与辐射场

$$\begin{aligned} A_z(\mathbf{x}) &= \frac{\mu_0 I_0 e^{ikR}}{2\pi k R} \frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin^2\theta} \\ \mathbf{B} = \nabla \times \mathbf{A} &= -i \frac{\mu_0 I_0 e^{ikR}}{2\pi R} \frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \mathbf{e}_\varphi \\ \mathbf{E} = c\mathbf{B} \times \mathbf{e}_r &= -i \frac{\mu_0 c I_0 e^{ikR}}{2\pi R} \frac{\cos\left(\frac{\pi}{2} \cos\theta\right)}{\sin\theta} \mathbf{e}_\theta \end{aligned}$$

半波天线平均能流密度

$$\mathbf{S} = \frac{1}{2} \operatorname{Re}(\mathbf{E} \times \mathbf{H}^*) = \frac{\mu_0 c I_0^2}{8\pi^2 R^2} \frac{\cos^2\left(\frac{\pi}{2} \cos\theta\right)}{\sin^2\theta} \mathbf{e}_r$$

半波天线辐射总功率

$$P = 2.44 \frac{\mu_0 c I_0^2}{8\pi} = \frac{1}{2} I_0^2 R_r$$

半波天线辐射电阻

$$R_r = 2.44 \frac{\mu_0 c}{4\pi} = 73.2 \Omega$$

8 狭义相对论

基本原理

1. 相对性原理:所有惯性参考系中物理定律的形式相同.
2. 光速不变原理:真空中的光速在任何惯性系中均为恒定值 c , 与光源运动无关.

Lorentz 变换

设 Σ' 系沿 Σ 系的 x 轴以速度 v 运动

$$\begin{cases} x' = \frac{x - vt}{\sqrt{1 - (v/c)^2}} \\ y' = y \\ z' = z \\ t' = \frac{t - xv/c^2}{\sqrt{1 - (v/c)^2}} \end{cases}$$

定义 $x_4 = ict$, 构成复四维空间. 令 $\gamma = 1/\sqrt{1 - \beta^2}$, $\beta = v/c$

$$\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \\ x'_4 \end{pmatrix} = \begin{pmatrix} \gamma & 0 & 0 & i\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -i\beta\gamma & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

速度变换

$$u'_x = \frac{u_x - v}{1 - \frac{v}{c^2}u_x} \quad u'_y = \frac{u_y}{\gamma \left(1 - \frac{vu_x}{c^2}\right)} \quad u'_z = \frac{u_z}{\gamma \left(1 - \frac{vu_x}{c^2}\right)}$$

运动尺度的缩短

$$l = \sqrt{1 - (v/c)^2} l_0 = l_0/\gamma$$

运动时钟的延缓

$$\Delta t = \frac{\Delta \tau}{\sqrt{1 - (v/c)^2}} = \gamma \Delta \tau$$

四维速度矢量

$$U_\mu = \frac{dx_\mu}{d\tau} = \gamma_u(\mathbf{u}, ic)$$

四维波矢量

$$k_\mu = \left(\mathbf{k}, i\frac{\omega}{c} \right)$$

相位 $\varphi = k_\mu x_\mu = k'_\nu x'_\nu$ 为 Lorentz 不变量.

四维波矢量变换

$$\begin{cases} k'_x = \gamma \left(k_x - \frac{v}{c^2} \omega \right) \\ k'_y = k_y \\ k'_z = k_z \\ \omega' = \gamma (\omega - vk_x) \end{cases}$$

横向 Doppler 效应

取 $\omega' = \omega_0$

$$\omega = \frac{\omega_0}{\gamma \left(1 - \frac{v}{c} \cos \theta \right)}$$

取 $\theta = \frac{\pi}{2}$

$$\omega = \frac{\omega_0}{\gamma}$$

四维电流密度矢量

$$J_\mu = \gamma_u(\rho_0 \mathbf{u}, ic\rho_0) = (\mathbf{J}, ic\rho)$$

四维势矢量

$$A_\mu = \left(\mathbf{A}, \frac{i}{c}\varphi \right) \quad (A'_\mu = a_{\mu\nu}A_\nu)$$

电磁场张量

$$F_{\mu\nu} = \frac{\partial A_\nu}{\partial x_\mu} - \frac{\partial A_\mu}{\partial x_\nu}$$

$$F_{\mu\nu} = \begin{pmatrix} 0 & B_z & -B_y & -iE_x/c \\ -B_z & 0 & B_x & -iE_y/c \\ B_y & -B_x & 0 & -iE_z/c \\ iE_x/c & iE_y/c & iE_z/c & 0 \end{pmatrix}$$

相对论电磁场变换公式

$$E'_x = E_x \quad E'_y = \gamma(E_y - vB_z) \quad E'_z = \gamma(E_z + vB_y)$$

$$B'_x = B_x \quad B'_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right) \quad B'_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right)$$

四维动量

$$p_\mu = mU_\mu$$

$$p_i = \gamma mv_i \quad (i = 1, 2, 3) \quad p_4 = ic\gamma m = \frac{i}{c} \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

物体总能量

$$W = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

动能

$$T = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2$$

相对论力学方程

$$K_\mu = \frac{dp_\mu}{d\tau} \quad K_\mu = \left(\mathbf{K}, \frac{i}{c} \mathbf{K} \cdot \mathbf{v} \right)$$

可分解为

$$\mathbf{K} = \frac{d\mathbf{p}}{d\tau} \quad \mathbf{K} \cdot \mathbf{v} = \frac{dW}{d\tau}$$

改为对 t 的微分

$$\sqrt{1 - \frac{v^2}{c^2}} \mathbf{K} = \frac{d\mathbf{p}}{dt} \quad \sqrt{1 - \frac{v^2}{c^2}} \mathbf{K} \cdot \mathbf{v} = \frac{dW}{dt}$$

定义力 $\mathbf{F} = \sqrt{1 - \frac{v^2}{c^2}} \mathbf{K}$, 则相对论力学方程化为

$$\begin{cases} \mathbf{F} = \frac{d\mathbf{p}}{dt} \\ \mathbf{F} \cdot \mathbf{v} = \frac{dW}{dt} \end{cases}$$